

How to setup windows managed host?

- you have a Linux Ansible control node and want to manage a Windows host.
- Since Ansible uses WinRM (not SSH) to talk to Windows, there are a few setup steps.

• Steps to Manage a Windows Host with Ansible

1. Prepare the Windows Managed Host

1. Enable PowerShell Remoting (WinRM)

On the Windows machine (run PowerShell as Administrator):

```
```bash
```

```
winrm quickconfig
```

```
```
```

2. Allow basic authentication & unencrypted traffic (for testing only)

```
```
```

```
winrm set winrm/config/service/auth @{Basic="true"}
```

```
winrm set winrm/config/service @{AllowUnencrypted="true"}
```

```
```
```

****OR****

```
```bash
```

```
Set-Item -Path WSMan:\localhost\Service\Auth\Basic -Value $true
```

```
Set-Item -Path WSMan:\localhost\Service\AllowUnencrypted -Value $true
```

```
```
```

⚠ For production, it's strongly recommended to use HTTPS with a certificate instead of unencrypted HTTP.

3. Check listening ports

...

```
netstat -an | findstr 5985
```

Port 5985 = HTTP

Port 5986 = HTTPS (needs certificate setup)

...

4. Create a dedicated Ansible user

- Add a local/domain user (e.g., ansible)
- Give it Administrator rights (or least privileges required).

...

```
New-LocalUser -Name "ansibleu" -Password (Read-Host -AsSecureString "Enter Password") -  
FullName "Ansible Automation User" -Description "User for Ansible automation"
```

```
Add-LocalGroupMember -Group "Administrators" -Member "ansibleu"
```

OR

```
net localgroup Administrators ansibleu /add
```

...

2. Install Required Packages on Control Node

On your Linux control node:

```
```bash
pip install pywinrm
```
```

```
```bash
python3 -c "import winrm; print(winrm.__version__)"
```
```

- Optional but recommended: install `ansible.windows` collection.

```
```bash
ansible-galaxy collection install ansible.windows
```
```

Ansible's Windows collection centralizes Windows modules and examples.

3. Configure Inventory for Windows

Edit your Ansible inventory file (/etc/ansible/hosts or custom):

```
...
```

```
[windows]
```

```
winhost ansible_host=192.168.244.56
```

```
[windows:vars]
```

```
ansible_user=ansibleu
```

```
ansible_password=yourwindowsuserpassword
```

```
ansible_connection=winrm
```

```
ansible_winrm_transport=basic
```

```
ansible_port=5985
```

```
ansible_become=no
```

```
...
```

For HTTPS + certificate auth, you'd use `ansible_winrm_transport=credssp` or `kerberos` instead.

4. Test the Connection

Run:

```
...
```

```
ansible windows -m win_ping
```

```
...
```

Expected output:

```
...
```

```
winhost | SUCCESS => {
```

```
  "changed": false,
```

```
  "ping": "pong"
```

```
}
```

5. Run a Sample Command

'''

```
ansible windows -m win_shell -a "ipconfig"
```

'''

Pro-Tips for Reliability:

- Use Ansible Vault to store passwords securely.
- Automate WinRM setup with a PowerShell bootstrap script so every new Windows VM is ready for Ansible.
- Prefer Windows-specific Ansible modules (, `win\_user`, `win\_package`) over raw shell commands for idempotency.

Thanks:

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